
Aprendizagem Computacional (AC) Machine Learning (ML)

2026/2027

Practical Exercises :
Week #1

Software Installation & Data Handling

Python Requirements

- Python 3
- Modules:
 - Numpy
 - Pandas
 - Sklearn
 - Stats
 - Plotly or Matplotlib
 - ...

1 Data handling

1.1 Excel data

1. Open the file *CORK_STOPPERS.XLS* in Excel, which is available at UCStudent-Folder Datasets. The file contains measurements performed automatically by an image processing system on 150 cork stoppers belonging to three classes. Each image was digitized with an adequate threshold to enhance the defects.

The first step is to read the available data description in the *Description* separator. The data description tells us that the dataset comprises a set of 10 discriminant features. Looking at the *Data* separator, we can observe that the columns C to L correspond to one feature. The A column enumerates the patterns, and the B column defines the class to which a given pattern belongs. Three categories are considered for the Cork Stoppers: 1-Quality Super; 2- Quality Average; 3- Quality Poor.

2. Make a scatter-plot representing the Cork Stoppers in a space generated by “N” and “ARM”? Can you visualize the three categories/classes?
3. Make a 3D scatter plot representing the Cork Stoppers in a data space spanned by features N, ARM, and PRM.

1.2 CSV data

1. Load the file *iris.csv* that contains 3 classes of 50 instances each, where each class refers to a type of iris plant. Each iris plant is represented by four features: “sepal_length”, “sepal_width”, “petal_length” and “petal_width”.
2. Make a scatter-plot representing the plants in a space spanned by “sepal_length”? Can you visualize the three categories/classes?
3. Make a scatter-plot representing the plants in a space spanned by “sepal_length” and by “petal_length”? Can you visualize the three categories/classes?
4. Make a scatter-plot representing the plants in a space spanned by “sepal_width”, “petal_length” and “petal_width”? Can you visualize the three categories/classes?